

QUERY/QUEST

Assignment 1:

Name: HIMNISH VERMA

Part I:

Q1

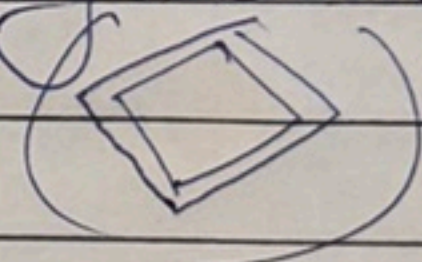
Ans. A traditional file system stores data in unrelated / isolated files, whereas a DBMS is a software that stores data in a structured way securely. An example where DBMS is superior is when you have made an order, but want to change the delivery address, in traditional file system, you have to update your address in multiple files but in DBMS, you only need to update in 1 table.

Q2

Ans Weak entity is entity with no key. Strong entity is an entity which has a key.

Primary key of weak entity is formed by combining the ^{key} of the strong entity with an attribute of weak entity.

The relationship is called identifying relationship, denoted by a double diamond in ER diagram.



Q3

Ans. It should be made as multivalued attribute for obvious reasons.

Q4

Ans. A candidate key of a relation is a minimal, non-primary key. A super key of a relation is the set of all possible keys of a relation.

In Student (roll-no, email, name, branch),

Candidate keys: {roll-no}, {email}.

not Candidate keys: {roll-no, email}, {email, branch}.

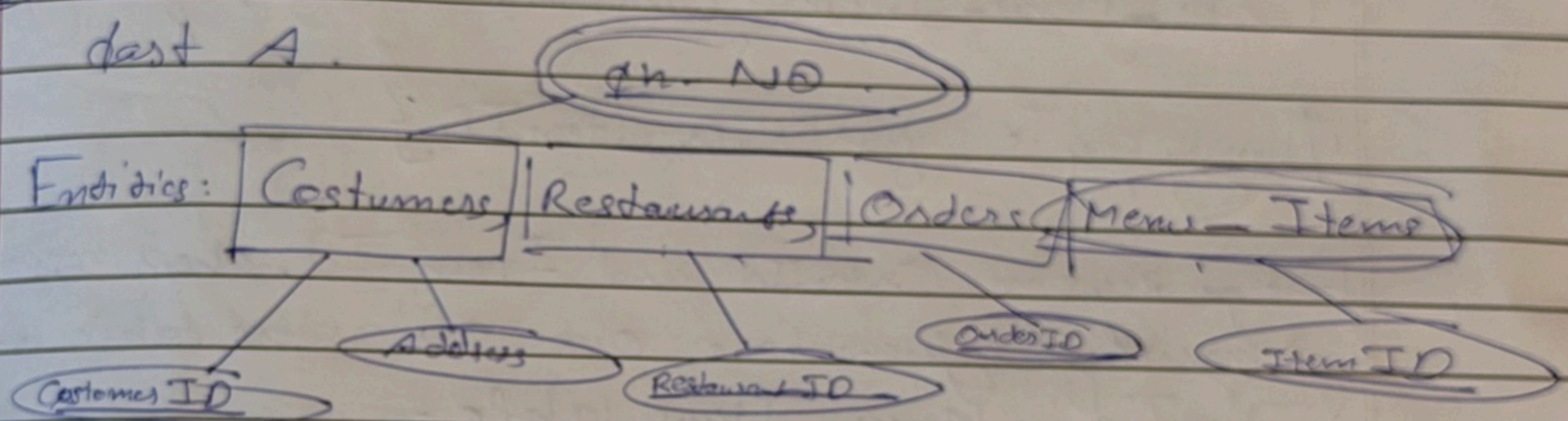
① Referential integrity is violated.

② It protects against data inconsistency.

③

Q. 1

Ans. Part A.



Customers - Orders is $1:N$, i.e. one customer can place any orders, but each order have only 1 customer.

Restaurants - Orders is $1:N$.

Restaurants - Menu-Items is $M:N$ as one restaurant can offer many items, and one item, like pizza, can be offered by multiple restaurants.

Orders - Menu-Items is also $M:N$.

Part B.

- Ph-No is multivalued, as customers can have multiple phone numbers registered.
- Address is composite attribute, as it requires information like street-name, house-no, city.

Part C:

Order-Menu-Items (OrderID, ItemID, quantity)

Primary key is {OrderID, ItemID}.

Foreign keys are OrderID and ItemID, their respective primary keys referenced are OrderID in orders table and ItemID in Menu-Items table.

Quantity is required. Quantity is a new attribute.

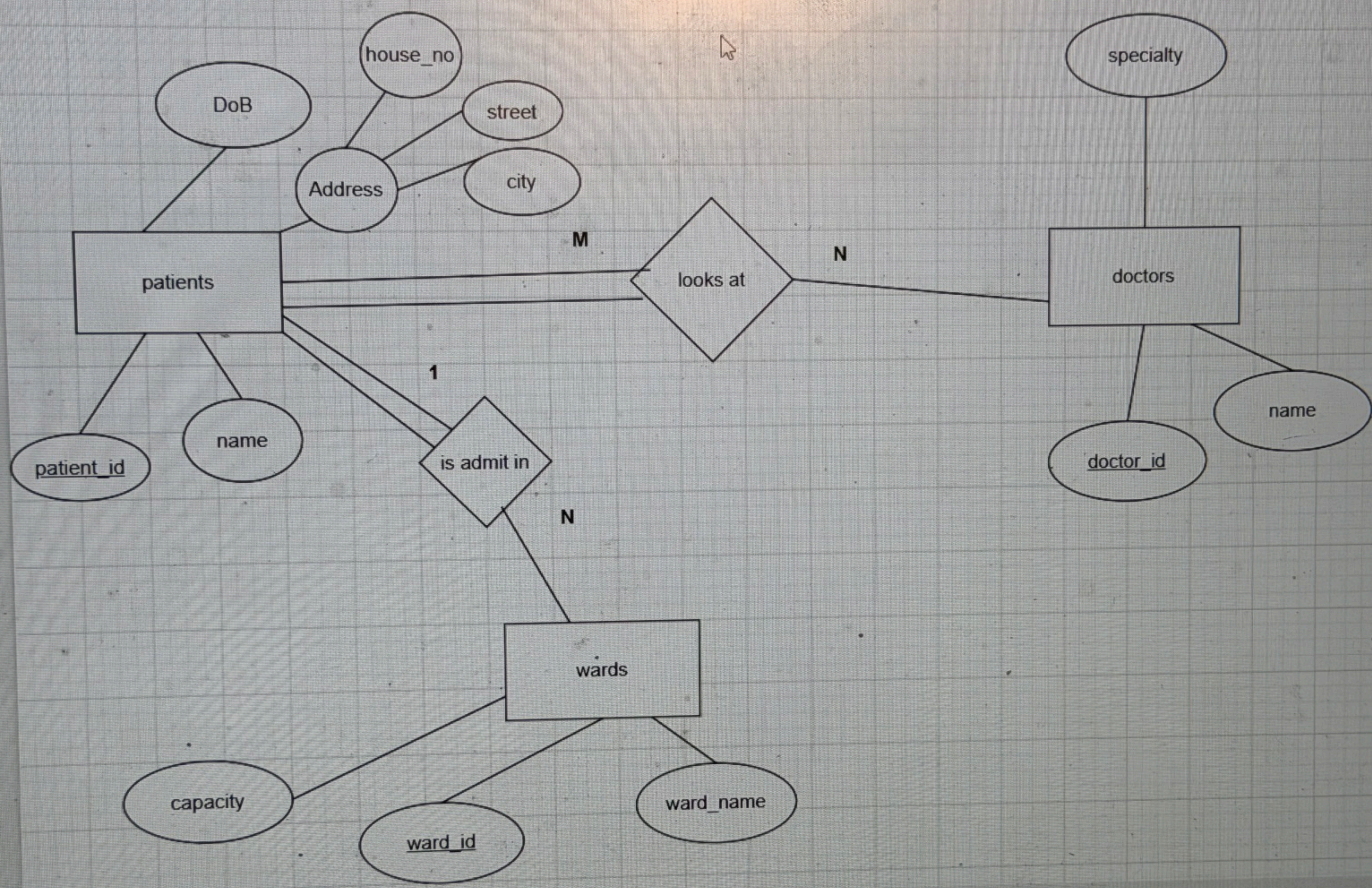
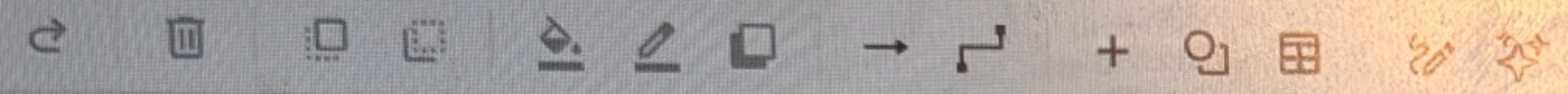
Part III

Q1

(On draw.io)

Q2

- The doctor-patient relationship is many-to-many. This is because a doctor can be looking at multiple patients simultaneously and a single patient may be going to multiple doctors for treatment of multiple diseases.



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- ☒ Connect
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Paper



This relation creates a new associated ^{entity} ~~entity~~, treatment. The composite primary key of this junction table is { doctor_id, patient_id }.

- In hospitals, a patient must always be admitted into a ward. So, there is total participation in patient entity set. The ward's entity set has partial participation, as it is not necessary that every ward should have a patient admitted, as some can be left empty. Doctors also need not have total participation, as some doctors may be on leave and not seeing a patient. But every patient must see a doctor, so there is total participation from the patient entity set.
- Address is a composite attribute, because address requires house-no, street and city. So, it has multiple subattributes. Hence, address is modelled as a composite attribute.